

A simple plotting example is shown in the following code block:

```
using Gadfly, RDatasets
# Load the Data Set
iris = dataset("datasets", "iris")
# Plot the Data
p = plot(iris, x=:SepalLength, y=:SepalWidth, Geom.point);
#Save it to SVG Image format with the specified dimensions
img = SVG("iris_plot.svg", 14cm, 8cm)
#Draw it on screen
draw(img, p)
```

The output of the code is shown in Figure 3.

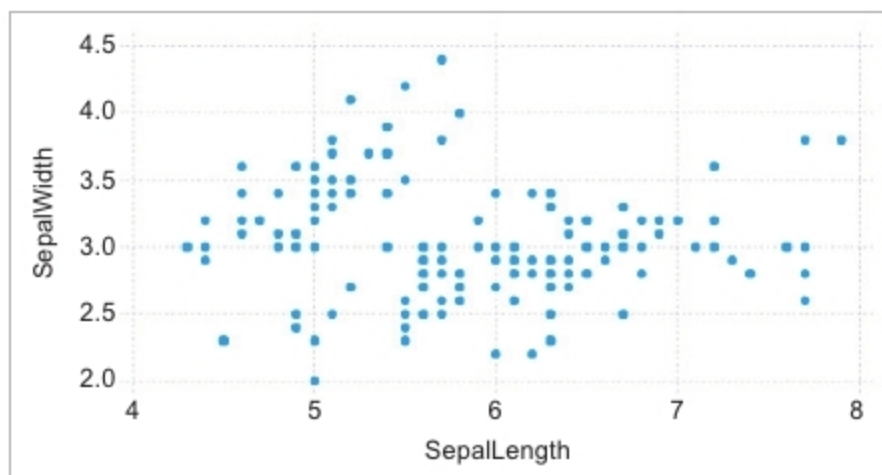


Figure 3: Gadfly plot

Gadfly has support for various types of charts. A complete tutorial on Gadfly is provided in the official documentation at <http://gadflyjl.org/stable/tutorial/#Tutorial-1>.

Makie

Makie is a powerful visualisation package available in Julia. To give readers some background information — Maki-e is the Japanese technique of spraying gold and silver powder. In today's cyber world, data is the gold dust. Makie enables the developer to use it to build stunning visualisation. The prominent features of *Makie.jl* are listed below:

- High performance
- The ability to extend
- Support for multiple platforms

Any package can be added to Julia with *Pkg.add*. The process of adding Makie is shown below:

```
Pkg.add("Makie")
```

The code to create a simple scatter plot is shown in the following code block:

```
using Makie
x = rand(10)
y = rand(10)
colors = rand(10)
```

```
scene = scatter(x, y, color = colors)
```

The output of the code is shown in Figure 4.

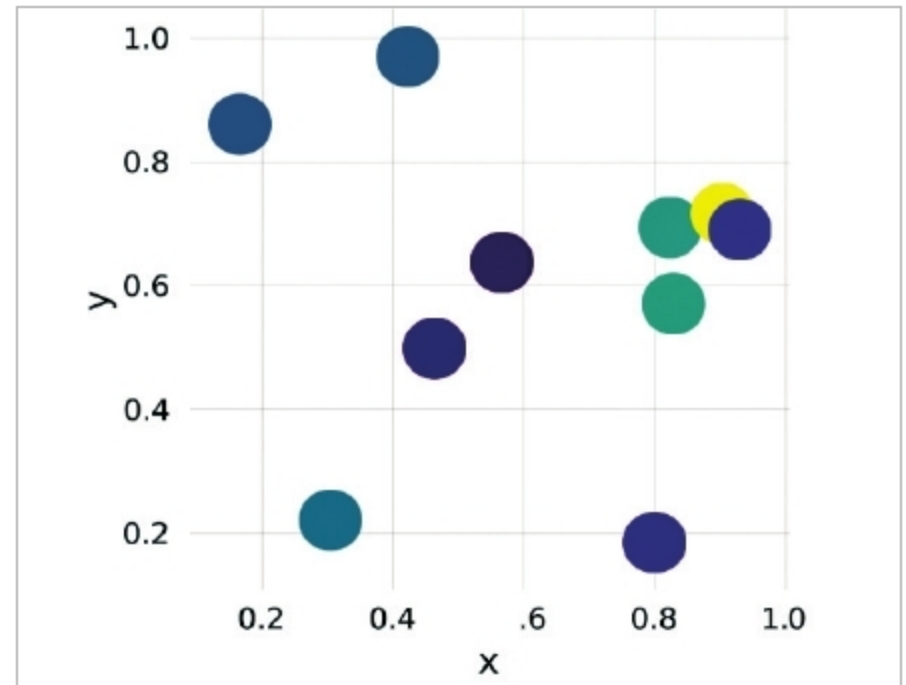


Figure 4: Makie scatter plot

The Makie code to build a line plot is shown in the following code block:

```
using Makie
x = range(0, stop = 2pi, length = 80)
f1(x) = sin.(x)
f2(x) = exp.(-x) .* cos.(2pi*x)
y1 = f1(x) y2 = f2(x)
scene = lines(x, y1, color = :blue)
scatter!(scene, x, y1, color = :red, markersize = 0.1)
lines!(scene, x, y2, color = :black)
scatter!(scene, x, y2, color = :green, marker = :utriangle,
markersize = 0.1)
```

The output of this code is shown in Figure 5.

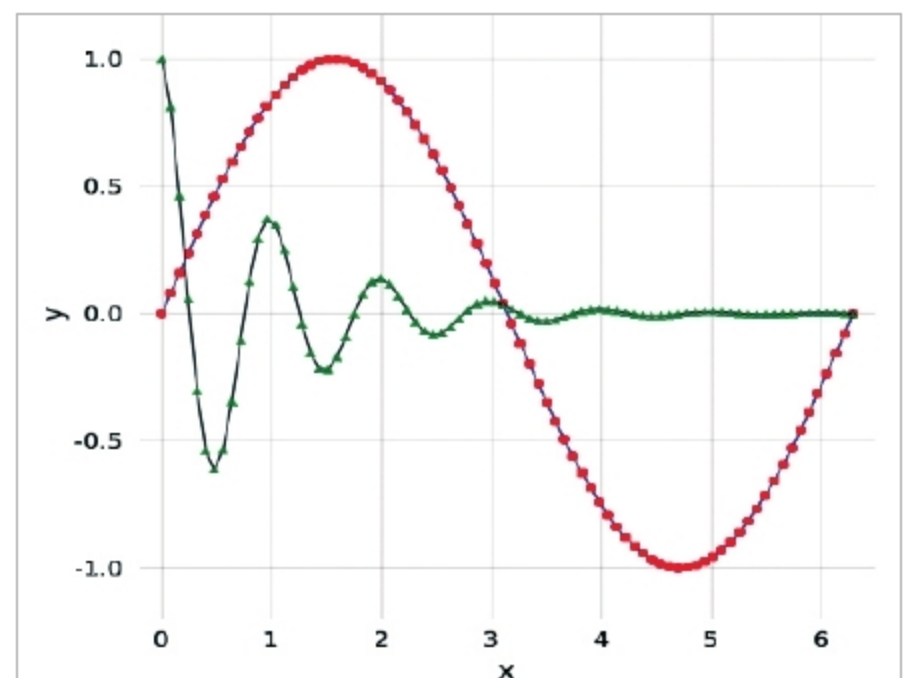


Figure 5: Makie line plot